

Workshop - Week 9

1. Evaluate the indefinite integral

(a) $\int x\sqrt{1-x^2}dx$

(b) $\int \frac{\cos \theta}{1+\sin \theta} d\theta$

(c) $\int \frac{a+bx^2}{\sqrt{3ax+bx^3}}dx$

(d) $\int \frac{\exp(\arcsin(x))}{\sqrt{1-x^2}}dx$

2. (a) If f is continuous on $\mathbb{R}$ prove that

$$\int_a^b f(x+c) = \int_{a+c}^{b+c} f(x)$$

(b) If a and b are positive numbers show that

$$\int_0^1 x^a(1-x)^b dx = \int_0^1 x^b(1-x)^a dx$$

3. Sketch the region enclosed by the given curves and find its area.

(a) $y = 12 - x^2$, $y = x^2 - 6$

(b) $x = 2y^2$, $x = 4 + y^2$

(c) $y = \cos(x)$, $y = 1 - \cos(x)$, $0 \leq x \leq \pi$

4. Find the volume of the solid obtained by rotating the region bounded by the given curves about the specified line. Sketch the region, the solid.

(a) $y = \exp(x)$, $y = 0$, $x = -1$, $x = 1$; about the x axis.

(b) $x = 2\sqrt{y}$, $x = 0$, $y = 9$; about the y axis.

5. First make a substitution and then use integration by parts to evaluate the integral.

(a) $\int_0^\pi \exp(\cos(t)) \sin(2t)dt$

(b) $\int x \ln(1+x)$

6. Evaluate the integral

(a) $\int \frac{x^2}{\sqrt{9-x^2}}$

(b) $\int_0^a x^2 \sqrt{a^2 - x^2} dx$